ADGNet: Attention Augmented Depthwise Separable Generative Adversarial Network for Image Dehazing

A Report submitted

in partial fulfillment for the Degree of

B. Tech.

in

Computer Science and Engineering

by

Aditya Shaurya Singh Negi
(221210012)

under the supervision of

Dr. Gautam Kumar
Assistant Professor

Department of Computer Science and Engineering

NATIONAL INSTITUTE OF TECHNOLOGY

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CERTIFICATE

This is to certify that the project report entitled **ADGNet: Attention Augmented Depthwise Separable Generative Adversarial Network for Image Dehazing** submitted by **Aditya Shaurya Singh Negi** to the National Institute of Technology, Delhi, in partial fulfillment for the award of the degree of **B. Tech. in Computer Science and Engineering** is a bona fide record of project work carried out by him under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

Dr. Gautam Kumar
Assistant Professor
Department of Computer Science and Engineering

DECLARATION

I declare that this project report titled **ADGNet: Attention Augmented Depthwise Separable Generative Adversarial Network for Image Dehazing**, submitted in partial fulfillment of the degree of **B. Tech. in Computer Science and Engineering**, is a record of original work carried out by me under the supervision of **Dr. Gautam Kumar**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

New Delhi
19 February 2026

Aditya Shaurya Singh Negi
221210012

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I also acknowledge the Department of Computer Science and Engineering, NIT Delhi, for providing the academic setting and resources needed to complete this project. I am grateful to the faculty members, friends, and classmates whose discussions, feedback, and encouragement helped me at different stages of the project.

A shortened version of this work has been published as the research paper *ADGNet: Attention Augmented Depthwise Separable Generative Adversarial Network for Image Dehazing*. This report develops the same work in greater detail by adding expanded background, implementation explanation, experimental discussion, and possible future extensions.

Aditya Shaurya Singh Negi

ABSTRACT

Outdoor visual data is often affected by haze, fog, or smoke. These conditions scatter light before it reaches the camera, which weakens contrast, changes color appearance, and hides scene information. The problem is particularly important in surveillance, emergency response, autonomous navigation, and other vision-based systems where unclear input can reduce the reliability of later decisions. The aim of this project is to improve the visibility of hazy images while keeping the model suitable for practical use.

This report presents ADGNet, a generative adversarial dehazing framework designed with an emphasis on efficiency. The generator uses an encoder–bottleneck–decoder arrangement with skip connections, while depthwise separable convolutions are introduced to lower the computational load. Residual blocks with squeeze-and-excitation attention are used in the central part of the network so that useful feature channels receive greater emphasis. A compact PatchGAN discriminator provides local adversarial feedback and helps the generator produce sharper and more natural-looking outputs.

Training and evaluation are performed on the Outdoor Training Set of the RESIDE benchmark, which contains paired hazy and clean images. The optimization objective combines adversarial supervision with pixel-wise reconstruction and an additional consistency term. The quality of restoration is measured using Peak Signal-to-Noise Ratio and Structural Similarity Index, while efficiency is examined through parameter count, FLOPs, and inference speed.

On the RESIDE OTS benchmark, ADGNet obtains a PSNR of 35.08 dB and an SSIM of 0.9722. The model has 1.60 million parameters, requires 15.74 GFLOPs for 256×256 images, and reaches 80.30 FPS on an NVIDIA T4 GPU. These results show that the proposed design provides a useful balance between image restoration accuracy and computational cost, making it relevant for real-time and resource-limited computer vision applications.

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ABBREVIATIONS/ NOTATIONS

ADGNet	Attention Augmented Depthwise Separable Generative Adversarial Network
AOD-Net	All-in-One Dehazing Network
BCE	Binary Cross Entropy
CNN	Convolutional Neural Network
DCP	Dark Channel Prior
DW	Depthwise
FPS	Frames Per Second
GAN	Generative Adversarial Network
GFLOPs	Giga Floating Point Operations
L1 Loss	Mean Absolute Error Loss
MSE	Mean Squared Error
OTS	Outdoor Training Set
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
RESIDE	Realistic Single Image Dehazing Dataset
SE Block	Squeeze-and-Excitation Block
SSIM	Structural Similarity Index Measure
Tanh	Hyperbolic Tangent Activation Function
U-Net	U-shaped Convolutional Network Architecture

CHAPTER 1

INTRODUCTION

1.1. Background

Vision-based systems are now used in many practical settings, including public surveillance, autonomous driving, robotics, traffic monitoring, and smart sensing platforms. Their performance depends on receiving images in which objects, boundaries, and scene details are visible. In outdoor environments, this requirement is often not satisfied because haze, fog, and smoke can reduce scene visibility before the image is even processed by an algorithm.

The visual degradation occurs because small particles in the air scatter light travelling from the scene to the camera. This scattering makes distant regions appear pale, lowers contrast, and reduces the separability of important objects from the background. Smoke creates a similar difficulty and is especially serious during fire incidents, where delayed recognition of people, exits, obstacles, or danger zones can affect safety-related decisions. For this reason, recovering a clearer image from a degraded input remains an important problem in computer vision.

Classical dehazing techniques generally relied on atmospheric image formation models and manually designed assumptions about natural images. Such techniques can work well in controlled cases, but their performance may fall when the scene does not satisfy the assumed prior. Deep learning changed this direction by allowing models to learn the relationship between degraded and clean images from data, making image dehazing a strong candidate for supervised restoration methods.

1.2. Motivation

Modern dehazing networks often report high benchmark scores, but many of them achieve this performance through deeper backbones, dense feature fusion, or costly attention modules. These choices increase memory use and inference time, which can make deployment difficult on cameras, drones, embedded processors, or edge devices.

The motivation behind this work is to design a model that treats efficiency as a primary requirement rather than an afterthought. A practical dehazing system should not only improve PSNR and SSIM, but should also run quickly, contain a manageable number of parameters, and remain stable during adversarial training. This project therefore focuses on a compact architecture that retains restoration quality while reducing unnecessary computational overhead.

1.3. Problem Statement and Objectives

The problem addressed in this project is to develop a deep learning framework that transforms hazy images into clearer outputs while preserving structural information and avoiding excessive computational complexity.

The main objectives are listed below:

- To build a GAN-based model for single image dehazing.
- To reduce computational cost by replacing selected standard convolutions with depthwise separable convolutions.
- To improve feature selection through channel-wise attention.
- To train and test the model on a recognized dehazing benchmark.
- To study the relationship among visual quality, inference speed, model size, and computational demand.

1.4. Overview of the Proposed Approach

The proposed method, ADGNet, is an attention-augmented depthwise separable GAN for image dehazing. It combines lightweight convolutional processing, residual squeeze-and-excitation attention, and PatchGAN supervision. Depthwise separable layers reduce the number of operations, SE blocks help the model prioritize important channels, and adversarial training encourages locally realistic image recovery.

The model is trained on paired hazy and clean images from RESIDE OTS. Restoration quality is measured using PSNR and SSIM. Efficiency is assessed using parameter count, FLOPs, and FPS. The design goal is to obtain competitive dehazing performance without depending on a very large network.

CHAPTER 2

LITERATURE REVIEW

Image dehazing has developed steadily because poor visibility affects both human interpretation and automated visual perception. Research in this area has moved from prior-based physical models to data-driven methods, including CNNs, attention-based networks, GANs, transformers, and hybrid models.

2.1. Traditional and Prior-Based Methods

Early methods mainly used the atmospheric scattering model along with assumptions about haze-free images. The Dark Channel Prior is one of the best-known examples. It is based on the observation that, in many clear outdoor patches, at least one color channel contains very low intensity values. This property can be used to estimate scene transmission and then reconstruct a less hazy image.

Although such methods provided an important starting point for dehazing research, their reliability depends on whether the underlying assumptions are true for the given image. They may struggle in sky regions, bright scenes, or images with unusual lighting. In some cases, they can also produce halos or unnatural colors. These weaknesses created interest in methods that learn restoration behavior directly from training data.

2.2. CNN-Based Dehazing Approaches

Convolutional neural networks brought a major shift by learning dehazing functions from paired examples. Models such as DehazeNet, AOD-Net, and MSCNN showed that a supervised network can estimate dehazed outputs more effectively than many manually designed approaches. Instead of explicitly estimating every physical component, these networks learn a mapping from hazy inputs to clean targets.

The strength of CNNs comes from their ability to form layered representations of an image. Shallow layers capture local textures and edges, while deeper layers can represent broader structures. However, early CNN dehazing models were sometimes limited when handling complex scenes or strong haze. This motivated later work on deeper designs, feature fusion, and multi-resolution processing.

2.3. Attention Mechanisms and Multi-Scale Networks

Attention and multi-scale strategies were introduced to help networks focus on the most useful information. Attention modules can assign higher importance to selected channels or spatial regions, while multi-scale processing allows the network to deal with haze patterns and objects of different sizes.

Methods such as GridDehazeNet, MSBDN, FFA-Net, and AECRNet improved restoration quality by using dense fusion, attention modules, or contrastive learning. Their results demonstrate the benefit of richer feature representation. At the same time, these improvements often increase parameter count, FLOPs, and implementation complexity, making real-time deployment harder.

2.4. GAN-Based Dehazing Methods

Generative adversarial networks approach dehazing from a perceptual viewpoint. A generator creates the restored image, and a discriminator judges whether image regions appear realistic. Through this competition, the generator is encouraged to recover sharper textures and more natural local detail than a purely pixel-wise loss may produce.

Models such as DW-GAN, Attentive GAN, and GAN-U-Net show the usefulness of adversarial learning for dehazing and desmoking tasks. However, GAN optimization can be sensitive to learning rates, loss balancing, and discriminator strength. In addition, several GAN-based dehazing systems use relatively heavy components, which can limit their suitability for low-resource environments.

2.5. Transformer and Hybrid Approaches

Transformer-based dehazing models have recently gained attention because self-attention can represent long-range dependencies across an image. This is useful when haze is spatially uneven or when distant regions influence the interpretation of the whole scene. Hybrid approaches combine physical imaging models with learned representations so that the method benefits from both domain knowledge and data-driven flexibility.

Despite their strong potential, many transformer and hybrid systems are computationally demanding. For applications such as surveillance, robotics, or embedded deployment, a model with smaller size and faster inference may be more useful even if the absolute benchmark score is slightly lower.

2.6. Research Gap

The literature shows continuous improvement in dehazing accuracy, but there remains a gap between high benchmark performance and practical deployment. Many methods focus mainly on maximizing quality scores and give less emphasis to speed, memory use, and training stability.

A deployable dehazing model should restore visual details, keep the architecture compact, train reliably, and process images at a practical rate. ADGNet is designed to address this need by combining depthwise separable convolution, residual channel attention, and a lightweight adversarial discriminator within a single framework.

CHAPTER 3

METHODOLOGY

This chapter explains the dataset, image preparation steps, model architecture, training procedure, loss design, stabilization methods, and evaluation criteria used for ADGNet.

3.1. Data Collection and Details

3.1.1. Dataset Description

The experiments are conducted using the Outdoor Training Set of RESIDE. RESIDE is widely used for single image dehazing research because it provides hazy images along with their clean reference images. The OTS portion contains more than 72,000 synthetic outdoor hazy samples created from clear images using atmospheric scattering. Since every hazy image has a matching ground truth image, the dataset supports supervised learning.

Figure 3.1: Hazy sample and matching clear reference image from the dataset.

3.1.2. Dataset Preprocessing

Before training, each image is resized to 256×256 pixels so that all samples have the same input size and the computational cost remains manageable. Pixel intensities are scaled to $[-1, 1]$ using a per-channel mean and standard deviation of 0.5, which matches the range produced by the Tanh activation at the generator output.

During training, random horizontal flipping is used as a light augmentation step. Stronger augmentations are avoided because they can alter the haze pattern and weaken the relationship between the hazy image and its clean target. The data is divided into training and validation subsets, and the validation portion is used to track restoration quality during training.

3.2. Proposed ADGNet Architecture

ADGNet follows a generative adversarial structure with two parts. The generator receives a hazy image and predicts a clearer version, while the discriminator examines local image regions and decides whether they resemble real clean-image patches. Figure 3.2 presents the overall design.

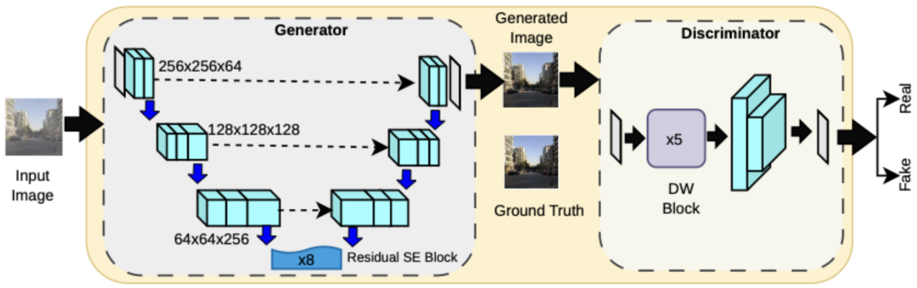

Figure 3.2: Overall architecture of ADGNet.

3.2.1. Depthwise Separable Convolution

In a normal convolution layer, spatial filtering and channel mixing are performed together. For a kernel of size D_k, C_{in} input channels, and C_{out} output channels, this requires $D_k \times D_k \times C_{in} \times C_{out}$ parameters. Depthwise separable convolution reduces this cost by splitting the operation into two simpler steps:

- **Depthwise convolution:** a separate spatial filter is applied to each input channel.
- **Pointwise convolution:** a 1×1 convolution then mixes information across channels.

This separation lowers the number of computations while still allowing the layer to extract spatial features and combine channel information.

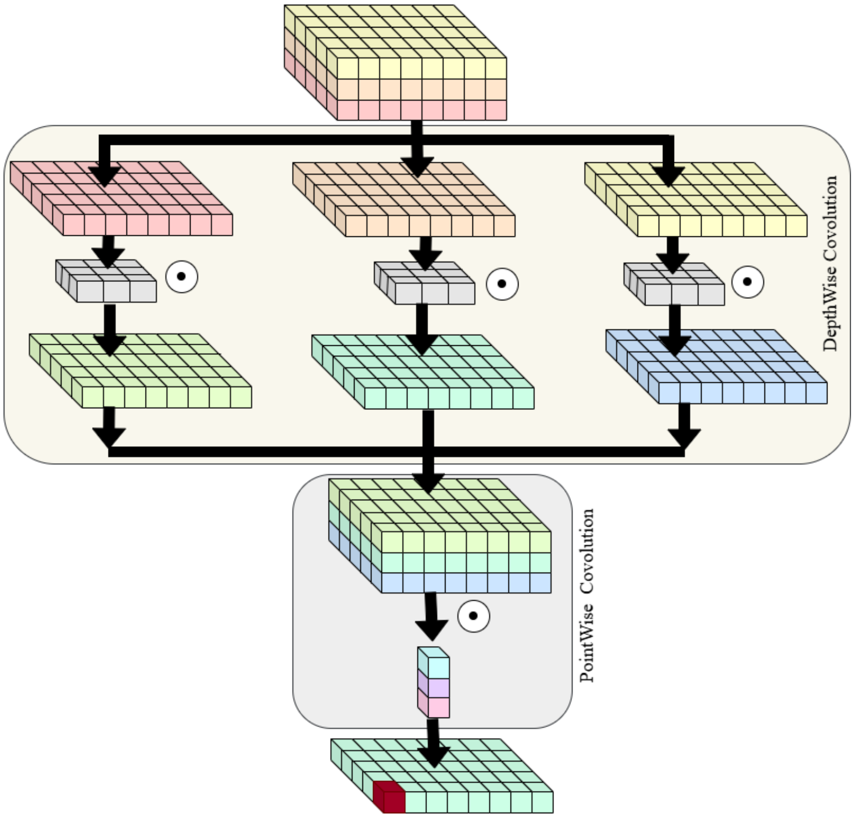

Figure 3.3: Depthwise and pointwise stages used in depthwise separable convolution.

3.2.2. Generator Network

The generator is organized as an encoder, a central bottleneck, and a decoder. Skip connections carry spatial information from the encoder to the decoder so that fine image details can be recovered during reconstruction.

Encoder

The first layer applies a 7×7 convolution with stride 1 and padding 3, converting the three-channel input into 64 feature maps. Instance normalization and ReLU follow this layer. The encoder then uses two stride-2 depthwise separable downsampling blocks, increasing the feature channels from 64 to 128 and then from 128 to 256. Normalization and LeakyReLU activation support stable feature extraction in these blocks.

Bottleneck

The bottleneck contains eight residual blocks with squeeze-and-excitation attention. Each block combines depthwise separable processing with a residual path. The SE component summarizes spatial information through global average pooling, learns channel weights using small fully connected layers, and rescales the feature maps. This helps the generator emphasize channels that are more useful for haze removal.

Decoder

The decoder converts encoded features back into an image. Two transposed convolution layers perform upsampling and reduce the number of channels from 256 to 128 and then from 128 to 64. Skip connections provide low-level details from the encoder. A final 7×7 convolution followed by Tanh produces a three-channel output image in the $[-1, 1]$ range.

3.2.3. Discriminator Network

The discriminator is based on PatchGAN. Instead of assigning one score to the whole image, it produces a $7 \times 7 \times 1$ map of local real-or-fake decisions. Each value corresponds to a local region, so the discriminator encourages the generator to improve textures, boundaries, and small-scale structures.

The discriminator begins with a convolutional layer and then uses five depthwise separable convolution blocks. Instance normalization, LeakyReLU, and dropout are included for stability and regularization. This design keeps the adversarial component lightweight while still providing useful feedback to the generator.

3.3. Simulation Setup

3.3.1. Hardware and Software Environment

The experiments are implemented in PyTorch. Training is carried out on an NVIDIA RTX 6000 Ada GPU. Inference speed is evaluated separately on an NVIDIA T4 GPU using images of size 256×256.

3.3.2. Training Configuration

The network is trained for 35 epochs with a batch size of 4. The Adam optimizer is used for both networks. The generator learning rate is 1×10^{-4}, and the discriminator learning rate is 4×10^{-4}. The Adam momentum terms are set to $\beta_1 = 0.5$ and $\beta_2 = 0.999$. A cosine annealing schedule is applied to adjust the learning rate smoothly over training.

3.3.3. Loss Function

The training loss combines adversarial learning, pixel-level reconstruction, and an additional image-consistency term:

$$L_{total} = L_{GAN} + \lambda_{L1} L_{L1} + \lambda_{perc} L_{perc} \quad (3.1)$$

Here, L_{GAN} guides the generator toward realistic image formation, L_{L1} reduces pixel-wise error with respect to the clean reference, and L_{perc} supports visually consistent reconstruction. The chosen weights are $\lambda_{L1} = 10$ and $\lambda_{perc} = 0.1$.

3.3.4. Training Stabilization Techniques

The following steps are used to make GAN training more stable:

- Real and fake labels are smoothed to 0.9 and 0.1 respectively.
- Generator and discriminator gradients are clipped with a maximum norm of 1.0.
- The generator is updated every alternate iteration.
- Instance normalization is used to improve image-to-image translation stability.
- Dropout is included in discriminator blocks to reduce overfitting.

3.3.5. Evaluation Metrics

Two standard image quality measures are used for evaluation: Peak Signal-to-Noise Ratio and Structural Similarity Index.

PSNR measures pixel-level fidelity and is computed as:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (3.2)$$

where MAX_I is the highest possible pixel value and MSE is the mean squared error between the generated image and its ground truth.

SSIM evaluates perceptual similarity by considering luminance, contrast, and structural information:

$$SSIM(x, y) = \frac{(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3.3)$$

where μ_x and μ_y denote means, σ_x^2 and σ_y^2 denote variances, σ_{xy} denotes covariance, and C_1 and C_2 prevent numerical instability.

CHAPTER 4

RESULTS AND ANALYSIS

4.1. Introduction

This chapter discusses the performance of ADGNet from five viewpoints: numerical restoration accuracy, visual quality, computational cost, ablation behavior, and robustness under different haze levels. All results are based on the RESIDE OTS evaluation setting described earlier.

4.2. Quantitative Performance Evaluation

Table 4.1 compares ADGNet with selected dehazing methods. The proposed model reaches 35.08 dB PSNR and 0.9722 SSIM, indicating that it reconstructs clean images with high pixel-level and structural similarity.

Table 4.1: Comparison of dehazing methods on RESIDE OTS.

Authors	Method	PSNR (dB)	SSIM
He et al.	DCP	16.62	0.8179
Li et al.	AOD-Net	19.82	0.8178
Cai et al.	DehazeNet	21.14	0.8472
Ren et al.	MSCNN	22.06	0.9078
Chen et al.	GCSNet	30.23	0.9800
Qin et al.	FFA-Net	36.39	0.9890
Fu et al.	DW-GAN	28.83	0.9330
Proposed Method	ADGNet	35.08	0.9722

The comparison shows that ADGNet performs much better than older prior-based and early CNN approaches. FFA-Net has a higher score, but it also uses a more complex attention-based structure. Therefore, the main advantage of ADGNet is not only its restoration score, but also the quality obtained relative to its compact design.

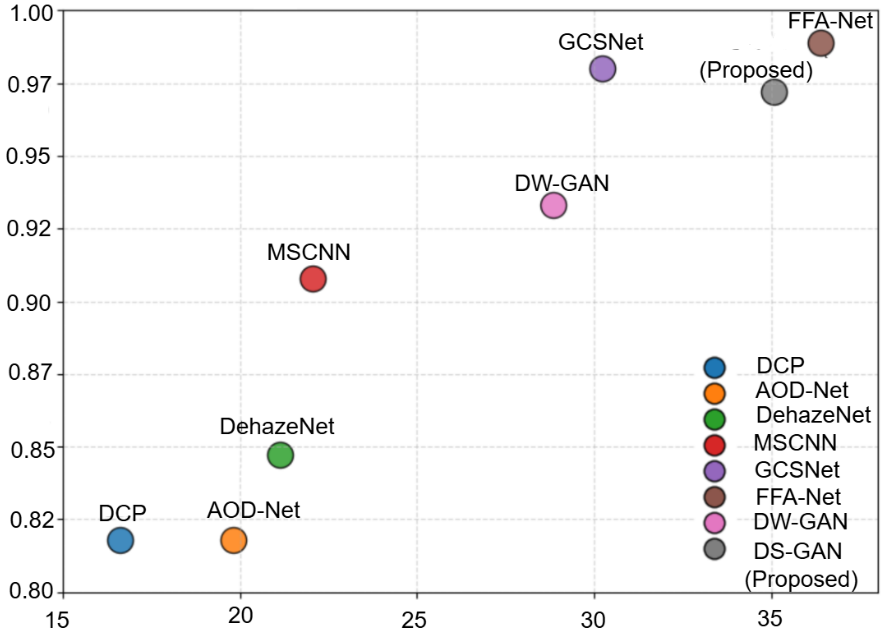

Figure 4.1: PSNR and SSIM comparison among selected dehazing methods.

On the validation set, the standard deviation is 2.67 dB for PSNR and 0.0166 for SSIM. These values suggest that the model does not perform well only on a few easy samples, but remains fairly consistent across different scenes and haze intensities.

4.3. Visual and Qualitative Analysis

Numerical metrics are useful, but visual inspection is also important because dehazing is ultimately an image restoration task. Figure 4.2 compares hazy inputs, ADGNet outputs, and clean references.

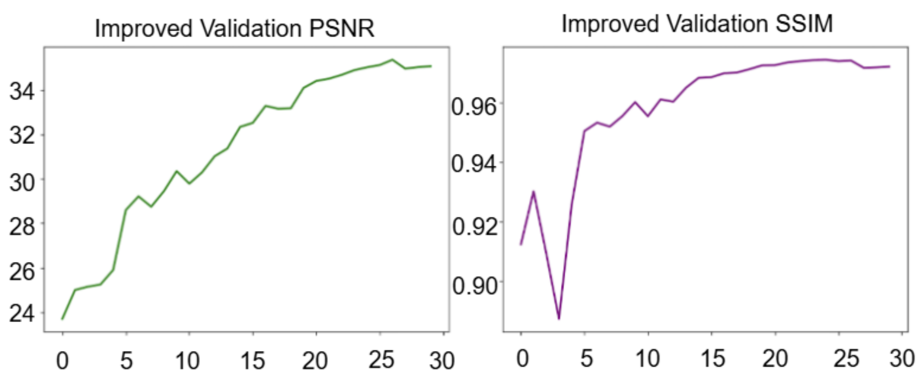

Figure 4.2: Visual comparison among input image, ADGNet output, and ground truth.

4.3.1. Natural Color Recovery

The restored images show more natural color balance than the hazy inputs. ADGNet improves brightness and removes the faded appearance caused by haze, while avoiding excessive saturation. This is important because aggressive color correction can make a dehazed image look artificial.

4.3.2. Structural Detail Preservation

ADGNet retains object boundaries, edges, and local structures in the output. Residual connections help information pass through the bottleneck, PatchGAN supervision encourages realistic local patterns, and SE attention helps the model select useful feature channels during reconstruction.

4.3.3. Improved Local Contrast

The outputs have stronger local contrast than the original hazy images. At the same time, the model does not introduce strong ringing or unnatural sharpening artifacts. This makes the restored images more useful for later tasks such as detection, tracking, or visual monitoring.

Figure 4.3: Validation-set distribution of PSNR and SSIM values.

4.4. Efficiency and Deployment Analysis

Efficiency is a central goal of this project. Table 4.2 compares model size, computational cost, and speed for ADGNet and representative baselines.

Table 4.2: Efficiency comparison of selected dehazing models.

Method	Params (M) ↓	FLOPs (G) ↓	FPS ↑
AOD-Net	0.80	8.40	90
FFA-Net	4.40	38.70	23
ADGNet	1.60	15.74	80.30

ADGNet contains 1.60 million parameters and requires 15.74 GFLOPs for a 256×256 input. It runs at 80.30 FPS on an NVIDIA T4 GPU. AOD-Net is smaller and faster, but its restoration quality is far lower. FFA-Net achieves stronger numerical performance, but at the cost of more parameters, higher FLOPs, and lower speed. ADGNet therefore occupies a useful middle position between accuracy and deployment efficiency.

4.5. Ablation Study

To examine the importance of depthwise separable convolutions, a baseline variant is created by replacing them with standard convolutions while leaving the rest of the setup unchanged.

Table 4.3: Effect of depthwise separable convolution in the ablation experiment.

Model	PSNR (dB)	SSIM
Standard convolution baseline	22.89	0.8421
ADGNet	35.08	0.9722

The baseline obtains 22.89 dB PSNR and 0.8421 SSIM, which is much lower than ADGNet. This should not be interpreted as proof that standard convolutions are always inferior. Instead, it shows that under the same training duration, learning rates, and hyperparameters, the depthwise separable version optimized more effectively and produced better features for this task.

4.6. Robustness Across Haze Intensities

The validation samples are grouped into light, medium, and heavy haze conditions to check whether performance remains stable as haze becomes stronger. Table 4.4 reports the results.

Table 4.4: ADGNet results for different haze-density groups.

Haze Density	PSNR (dB)	SSIM
Light	36.20	0.9835
Medium	35.08	0.9722
Heavy	33.72	0.9583

As expected, performance decreases when haze becomes heavier, but the drop is gradual. Even in the heavy-haze group, PSNR stays above 33 dB and SSIM remains higher than 0.95. This indicates that the architecture is not limited to only lightly degraded images and can handle a range of atmospheric conditions.

4.7. Overall Discussion

The results support the main design objective of ADGNet: balancing restoration quality with computational practicality. Depthwise separable convolutions reduce the cost of feature extraction, SE blocks improve channel selection, and adversarial learning improves local realism. Although larger networks can sometimes obtain better peak benchmark values, ADGNet provides a more deployment-friendly option for real-time dehazing tasks.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1. Conclusion

This project developed ADGNet, a compact GAN-based approach for single image dehazing. The model combines depthwise separable convolution, residual squeeze-and-excitation attention, and a lightweight PatchGAN discriminator to restore clearer images from hazy inputs without making the network unnecessarily heavy.

Evaluation on RESIDE OTS shows that ADGNet achieves 35.08 dB PSNR and 0.9722 SSIM. It uses 1.60 million parameters, requires 15.74 GFLOPs, and reaches 80.30 FPS on an NVIDIA T4 GPU. These results show that the method provides competitive restoration performance while remaining suitable for real-time use.

The ablation results indicate that the depthwise separable design contributes not only to efficiency but also to stable optimization in the chosen training setup. The robustness analysis further shows that the model maintains good performance across light, medium, and heavy haze categories. Overall, the project shows that lightweight convolutional design and channel attention can be combined effectively for practical image dehazing.

5.2. Future Work

Future improvements can proceed in several directions. The first direction is to evaluate the model on real hazy and smoky scenes, since real atmospheric effects can differ from synthetic haze. The second direction is video dehazing, where temporal consistency between consecutive frames would be important. The third direction is deployment-oriented optimization, such as quantization, TensorRT conversion, or testing on embedded hardware. Finally, transformer-based modules or hybrid physical-learning designs may be explored for scenes with complex and non-uniform haze.

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